CSA0919 PROGRAMMING IN JAVA

STRING PROGRAMS 1-14

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1

CharFrequency.java

```java
import java.util.*;

public class CharFrequency {
    public static void main(String[] args) {
        String str = "program";
        int[] freq = new int[256];

        for (char c : str.toCharArray())
            freq[c]++;

        for (char c : str.toCharArray())
            if (freq[c] != 0) {
                System.out.println(c + " : " + freq[c]);
                freq[c] = 0;
            }
    }
}
```

Output

```
p : 1
r : 2
o : 1
g : 1
a : 1
m : 1

=== Code Execution Successful ===
```

2

StringLength.java

```java
public class StringLength {
    public static void main(String[] args) {
        String str = "Hello";
        System.out.println("Length: " + str.length());
    }
}
```

Output

```
Length: 5

=== Code Execution Successful ===
```

3

CompareStrings.java

```java
import java.util.Scanner;

public class CompareStrings {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first string: ");
        String s1 = sc.nextLine();

        System.out.print("Enter second string: ");
        String s2 = sc.nextLine();

        if(s1.equals(s2))
            System.out.println("Strings are equal");
        else
            System.out.println("Strings are not equal");
    }
}
```

Output

```
Enter first string: java
Enter second string: java
Strings are equal

=== Code Execution Successful ===
```

4

ReverseString.java

```java
import java.util.Scanner;

public class ReverseString {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        String rev = "";

        for(int i=str.length()-1;i>=0;i--){
            rev = rev + str.charAt(i);
        }

        System.out.println("Reversed string: " + rev);
    }
}
```

Output

```
Enter a string: rose
Reversed string: esor

=== Code Execution Successful ===
```

5

ConcatStrings.java

```java
import java.util.Scanner;

public class ConcatStrings {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first string: ");
        String s1 = sc.nextLine();

        System.out.print("Enter second string: ");
        String s2 = sc.nextLine();

        String result = s1 + s2;

        System.out.println("Concatenated String: " + result);
    }
}
```

Output

```
Enter first string: hello
Enter second string: world
Concatenated String: helloworld

=== Code Execution Successful ===
```

6

CopyString.java

```java
import java.util.Scanner;

public class CopyString {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String s1 = sc.nextLine();

        String s2 = s1;

        System.out.println("Copied string: " + s2);
    }
}
```

Output

```
Enter a string: java
Copied string: java

=== Code Execution Successful ===
```

7

CountCharacters.java

```java
        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        for(int i=0;i<str.length();i++){
            char ch = str.charAt(i);

            if("aeiouAEIOU".indexOf(ch)!=-1)
                vowels++;
            else if(Character.isLetter(ch))
                consonants++;
            else if(Character.isDigit(ch))
                digits++;
            else if(ch==' ')
                spaces++;
        }

        System.out.println("Vowels: " + vowels);
        System.out.println("Consonants: " + consonants);
        System.out.println("Digits: " + digits);
        System.out.println("Spaces: " + spaces);
    }
}
```

Output

```
Enter a string: hi i am rose
Vowels: 5
Consonants: 4
Digits: 0
Spaces: 3

=== Code Execution Successful ===
```

8

CharacterType.java

```java
import java.util.Scanner;

public class CharacterType {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);

        if(Character.isUpperCase(ch))
            System.out.println("Uppercase Letter");
        else if(Character.isLowerCase(ch))
            System.out.println("Lowercase Letter");
        else if(Character.isDigit(ch))
            System.out.println("Digit");
        else
            System.out.println("Special Character");
    }
}
```

Output

```
Enter a character: Jrose4
Uppercase Letter

=== Code Execution Successful ===
```

9

SortStrings.java

```java
import java.util.Scanner;
import java.util.Arrays;
public class SortStrings {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of strings: ");
        int n = sc.nextInt();
        sc.nextLine();

        String[] arr = new String[n];

        for(int i=0;i<n;i++){
            System.out.print("Enter string: ");
            arr[i] = sc.nextLine();
        }

        Arrays.sort(arr);

        System.out.println("Sorted Strings:");
        for(String s: arr)
            System.out.println(s);
    }
```

Output

```
Enter number of strings: 3
Enter string: sun
Enter string: rose
Enter string: hair
Sorted Strings:
hair
rose
sun

=== Code Execution Successful ===
```

10

SwapStrings.java

```java
import java.util.Scanner;

public class SwapStrings {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first string: ");
        String s1 = sc.nextLine();

        System.out.print("Enter second string: ");
        String s2 = sc.nextLine();

        String temp = s1;
        s1 = s2;
        s2 = temp;

        System.out.println("After Swapping:");
        System.out.println("First String: " + s1);
        System.out.println("Second String: " + s2);
    }
}
```

Output

```
Enter first string: nav
Enter second string: niv
After Swapping:
First String: niv
Second String: nav

=== Code Execution Successful ===
```

11

CountWords.java

```java
import java.util.Scanner;

public class CountWords {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a sentence: ");
        String str = sc.nextLine();

        String[] words = str.split(" ");

        System.out.println("Number of words: " + words.length);
    }
}
```

Output

```
Enter a sentence: rose is red
Number of words: 3

=== Code Execution Successful ===
```

12

FindSubstring.java

```java
import java.util.Scanner;

public class FindSubstring {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter main string: ");
        String str = sc.nextLine();

        System.out.print("Enter substring: ");
        String sub = sc.nextLine();

        if(str.contains(sub))
            System.out.println("Substring found");
        else
            System.out.println("Substring not found");
    }
}
```

Output

```
Enter main string: rose
Enter substring: lilly
Substring not found

=== Code Execution Successful ===
```

13

PalindromeString.java

```java
import java.util.Scanner;
public class PalindromeString {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String str = sc.nextLine();
        String rev = "";
        for(int i=str.length()-1;i>=0;i--){
            rev = rev + str.charAt(i);
        }
        if(str.equals(rev))
            System.out.println("String is Palindrome");
        else
            System.out.println("String is not Palindrome");
    }
}
```

Output

```
Enter a string: jose
String is not Palindrome

=== Code Execution Successful ===
```

14

VowelOrConsonant.java

```java
import java.util.Scanner;

public class VowelOrConsonant {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);

        if("aeiouAEIOU".indexOf(ch)!=-1)
            System.out.println("It is a Vowel");
        else
            System.out.println("It is a Consonant");
    }
}
```

Output

```
Enter a character: vowel
It is a Consonant

=== Code Execution Successful ===
```